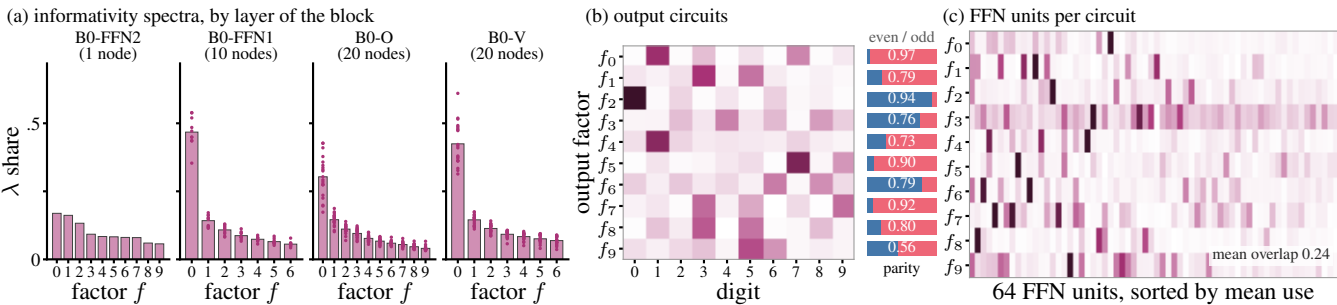


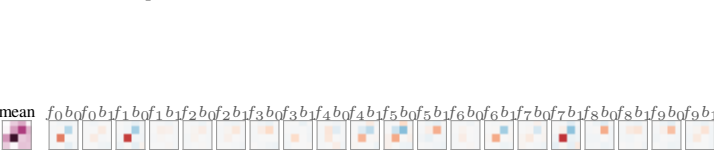


(d) L_{FFN1} sub-factors of each output circuit — weighted-average input (top, bold: dominant digit) and per-digit loading (bottom; blue even, red odd)

f_0 1/6 1/6 3/3 1/6 f_1 3/2 3/5 5/3 2/5 f_2 0/2 0/0 0/0 0/0 f_3 4/8 4/9 4/8 2/6 f_4 1/6 2/6 1/6 1/6 f_5 7/8 7/12 5/3 1/6 f_6 6/8 4/9 6/8 6/6 f_7 9/8 9/9 9/5 9/6 f_8 3/8 5/5 3/3 3/6 f_9 5/6 5/5 5/5 5/5



(e) CLS attention per factor



CLS-token weight 0.026; deviation of each leaf k_0 from the mean
(all 139 factor maps: pairwise cos ≥ 0.93)

(f) strongest stimuli per circuit

